

A Practical Feeder Planning Model for Urban Distribution System

Mingqiang Wang , Member, IEEE, Ming Yang , Senior Member, IEEE, Zhen Fang, Mengxia Wang , Member, IEEE, and Qiuwei Wu , Senior Member, IEEE

Abstract—There is a significant gap between academic research and practical application for power distribution system planning (PDSP). For most of the existing PDSP models in academic research, cost is used as the objective function, and the most common constraints are power flow equality constraints, bus voltage or voltage drop limits, substation and feeder capacity limits, etc. Although various advanced models and methods have been proposed, they are rarely used in real distribution system companies. This paper proposes a new feeder planning model for the urban distribution network considering various practical requirements. According to field investigation, “supply electricity to loads using nearby power sources” is used as the objective, and it is analytically expressed by load moment. Block loads, practical street layout, meshed network configuration, connection mode of feeders, non-crossing requirement of feeders, power supply radius, etc., are all respected and embedded in the model. The proposed model is recast as a mixed integer linear programming (MILP) problem, and it can be solved by the state-of-art commercial solver. The effectiveness and validity of the proposed model are illustrated on a 9-block test system and a real 22-block district distribution system.

Index Terms—Block loads, connection mode, mixed integer linear programming, street layout, urban distribution network planning.

NOMENCLATURE

A. Indices

b	index of blocks.
i, j, k	index of junctions.
$l, l1, l2$	index of feeders or distribution lines.
s	index of segments; and
g	index of substations.

B. Variables

1) Binary Variables

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Mingqiang Wang, Ming Yang, Zhen Fang, and Mengxia Wang are with the Key Laboratory of Power System Intelligent Dispatch and Control, Ministry of Education, Shandong University, Jinan, Shandong 250002, China (e-mail: wang0367@sdu.edu.cn; myang@sdu.edu.cn; dzwbzsfz@163.com; wangmx@sdu.edu.cn).

Qiuwei Wu is with the Department of Electrical Engineering, Technical University of Denmark, 2800 Kgs. Lyngby, Denmark (e-mail: qw@elektro.dtu.dk).

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$b_{s,l}$

a binary variable to represent whether feeder l is constructed in segment s . If feeder l is constructed in segment s , $b_{s,l}$ takes 1, otherwise it takes 0.

2) Continuous Variables

$D_{b,s}^{segment}$

equivalent demand transferred from block b to segment s .

$D_{b,s,l}^{segment}$

equivalent demand transferred from block b to feeder l of segment s .

$P_{i,j,l}$

power flows from junction i to junction j in feeder l ; and

$P_{i,l}^G$

power flows in feeder l and it flows from substation which is connected to junction i .

C. Constants

D_b^{block}

equivalent demand in block b .

$H_{b,s}$

relationship between block b and segment s . If segment s surrounds block b , $H_{b,s}$ takes 1, otherwise it takes 0.

$K_{s,j}^4$

a parameter to represent the connection relationship between segment s and junction j which connects with four segments.

L_{max_radius}

maximum power supply radius.

L_s

length of segment s .

$L_g^{location}$

junction where substation g is located.

N_j^{maxGL}

maximum number of outgoing feeders of the substation which is connected to junction j .

N_j^{minGL}

minimum number of outgoing feeders of the substation which is connected to junction j .

N_s^{maxL}

maximum number of feeders allowed to be built in segment s .

NB

total number of blocks.

NJ

total number of junctions.

NL_s

maximum number of feeders allowed to be built in segment s .

NS

total number of segments.

P_j^{Gmax}

maximum allowable output power of substation which is connected to junction j .

$P_{s,l}^{max}$

maximum allowable power flows in feeder l in segment s .

η_i^G

utilization factor of substation which is connected to junction i ; and

η_l^L

utilization factor of feeder l which is assigned by planning experts.

I. INTRODUCTION

POWER distribution system planning (PDSP) is an important subject of power system planning, and it has a significant impact on the economic and reliable operation of distribution systems [1]. As an important content of the PDSP, feeder planning [2], [3], which determines the optimal routing and sizing of feeders, has direct effects on the reliability of power supply to users, and becomes urgent to meet the load growth in the future.

Most of the existing research related to PDSP minimizes the net present value of the costs, which includes investment cost, or investment cost plus energy loss cost, or investment cost plus energy loss cost and reliability cost, or investment, loss, reliability, operation and maintenance cost, or total costs minus revenues. The common constraints include power flow constraints, bus voltage or voltage drop limits, substation and feeder capacity limits, radial operation of network, and full connectivity, etc. The PDSP model can be solved by various numerical methods or heuristic methods. The detailed comprehensive overview of the models and methods of PDSP can be found in [4], [5].

Although various advanced models and methods have been proposed, there is a lack of understanding about PDSP from a practical point of view [6]. Since design criteria and assumptions are rarely close to the industry practices [7], most of the research work has only academic interest and little has been applied in reality [6], [8]. The majority of distribution companies still use heuristic processes and empirical rules through expert judgement and practical analysis for PDSP. The collaboration between researchers and utility planners provides great opportunities for the development of models and methods well fitted to the utility needs and applied for real-world PDSP problems of distribution companies [4].

This paper focuses on the single stage feeder planning of medium voltage (MV) urban distribution system from scratch, and various practical requirements from utility planners based on company-specific policies are well respected. According to the on-site investigation, it is found that the utility planners have the following practical concerns. 1) The objective is “supply electricity to loads using nearby power sources”. 2) Block loads are applied in urban areas and every block load must be supplied by electricity. 3) The feeders should be alongside the streets, or the practical street layout should be considered. 4) The principle of “closed loop design, open loop operation”, or meshed network configuration, must be followed. 5) The feeder connection mode must be respected. 6) The feeders cannot be crossed. 7) Power supply radius should be considered. 8) Other specific constraints, such as maximum number of outgoing feeders of substations, capacity limits of substations and feeders, etc., must be satisfied.

For the objective function is concerned, nearly all the existing research [4] uses various costs as their objective function. However, it is found that the utility planners are reluctant to use cost as the objective function. The main reasons are as follows. In academic research, it assumes that all input data are deterministic and accurate. Various uncertainties are very hard (or even infeasible) to be modeled [8], and significant assumptions and simplifications are applied when calculating

the objective function. For example, when the PDSP is based on a typical day, the yearly operating cost is approximately computed by multiplying the operating cost of the typical day with 365, and the same is true for calculating the reliability cost. The effects of these assumptions and simplifications on the optimization results are not well analyzed, and the results are not very convincing. Therefore, the utility planners propose the objective of “supply electricity to loads using nearby power sources”. This objective is more straightforward and easy to understand, and it is consistent with “minimizing investment cost and operation cost” in essence. “Supply electricity to loads using nearby power sources” implies that the feeders need to be constructed should be relatively short, and the investment cost tends to be minimized. “Supply electricity to loads using nearby power sources” implies that the electrical distances between sources and loads are relatively short, and the loss or operation cost tends to be minimized. However, this objective is a qualitative expression and it needs to be explicitly and analytically represented.

Block loads and street layout are not widely considered in the PDSP. Conventionally, nodal loads and power sources are applied, and the feeder planning problem belongs to a node-to-node connection problem. Under this condition, the results obtained are inaccurate and may lose plasticity [9]. In reality, it is observed that feeders are usually alongside the streets in urban areas. The feeder planning problem can be represented in more details by involving the street layout, and the planning results would be more straightforward for practical application. There is little work [9]–[13] considering the practical street layout in the PDSP. Reference [9] optimizes the locations and capacities of substations and feeders for urban distribution system considering the street layout. The planning problem is analytically formulated as a mixed integer linear programming (MILP) model. Reference [10] generates the feeder map for reference distribution network considering the street layout. The feeder map is recognized by a heuristic method based on the Delaunay triangulation under some given principles. Reference [11] presents a planning method for low-voltage distribution networks considering the street layout. The network layout is determined by a heuristic method based on the minimum expansion tree. Reference [12] proposes a PDSP model based on given feeder corridors which usually take the same shape as the street layout. Reference [13] considers the street layout during the feeder block division procedure. For the loads are concerned, nearly all the researches apply the nodal load points. The block loads, i.e., loads in the unit of real block, are only applied in paper [9].

Most of the PDSP researches consider the radial network configuration [1], which are typically not applicable to real distribution systems [14]. A mesh-constructed network structure includes redundant tie-lines which are organized as backups for rerouting power in the event of feeder failures, and thus is more reliable than a radial-constructed network [15]. There are just a few references [6], [12], [14]–[23] considering the meshed network configuration in the PDSP. References [12], [14]–[17] consider the meshed network configuration by explicitly considering the post-fault load restoration. References [6],

[7], [18]–[20] consider the meshed network configuration by explicitly considering the 2-step ladder layout [7], [18], H-shaped layout [6], loop [19], and multi-loop connection modes [20] of feeders. References [21]–[23] consider the meshed network configuration by a two-step method, i.e., the radial distribution networks are planned first, then the terminal nodes of feeders are interconnected in order to generate a close-loop network configuration.

The connection modes of feeders directly affect the results of the PDSP [22]–[23]. In reality, the connection mode is usually considered since it can explicitly guarantee the reliability level of distribution systems. For unification, usually only one connection mode is considered in a distribution system. The standardization of connection mode makes the network easier to be operated, reducing operational mistakes, and improving maintenance and repairing scheduling [24]. There are just a few references [6], [7], [18]–[23] considering the connection mode of feeders in the PDSP. References [6], [7], [18] optimally integrate two separate MV distribution networks in Milano, Italy, by exploiting the 2-step ladder and H-shaped layouts. Reference [19] proposes an evolutionary algorithm for the planning of MV urban distribution networks considering the loop connection mode. Reference [20] designs the MV network considering the multi-loop connection mode using the combination of the ant colony system and genetic algorithm. Reference [21] proposes an urban MV distribution network planning model considering the single-ring connection mode based on the adaptive simulated annealing genetic algorithm and path optimization method for the travelling salesman problem. Reference [22] proposes a double-Q planning model for large-scale MV distribution networks considering the connection mode constraints. Reference [23] presents a method for optimal planning of new urban distribution networks based on the selection of the best subset of paths with back feed from the entire path set, and interconnecting and ring connection modes are considered in the model.

However, the existing models are represented with less detail than that required by utility planners in reality. Block loads, street layout, meshed network configuration, and feeder connection mode are not well respected simultaneously in the existing PDSP models. Besides, according to the on-site investigation, the utility planners strongly argue that the feeders cannot be crossed. This is mainly caused by the necessity for partition management. So far as our knowledge goes, the non-crossing constraint of feeders is not respected in existing PDSP models.

In this paper, a new feeder planning model for urban distribution system is proposed considering various practical requirements proposed by utility planners. According to field investigation, “supply electricity to loads using nearby power sources” is used as the objective, and this objective is analytically expressed by minimizing the load moment, which can be formulated as the summation of the product of feeder length and absolute value of power flow on the feeder. Various practical constraints are considered, such as consumers are seen as block loads rather than nodal loads, practical street layout, meshed network configuration, connection mode of feeders, non-crossing constraint of feeders, power supply radius, etc. And these requirements have been directly embedded into the

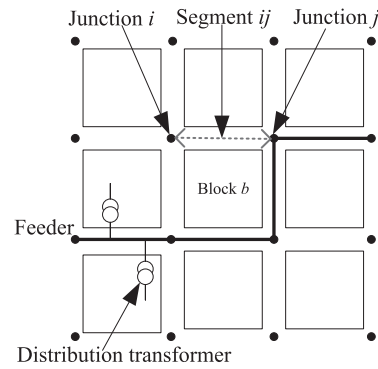


Fig. 1. An illustrative example to show blocks, junctions, and segments.

proposed model. In order to improve the computational efficiency, the model is recast as a MILP problem and it could be effectively solved by state-of-art commercial solvers. The effectiveness and validity of the proposed model are illustrated based on a hypothetical 9-block system and a real 22-block district distribution system.

It should be emphasized that among all the input data of the proposed model, only the block loads are forecast values and they are uncertain. The other input data are all deterministic. The proposed model is more robust than the existing PDSP models in terms of input data required.

The major contributions of the paper are as follows.

- 1) A new feeder planning model for urban distribution system is proposed. Load moment is used as the objective function. The block load, practical street layout, meshed network configuration, feeder connection mode, and non-crossing constraint of feeders are all respected in the model.
- 2) The proposed model is recast as a MILP problem and it can be effectively solved by state-of-art commercial solvers.

The rest of the paper is organized as follows. In section III, the proposed feeder planning model is proposed. Section IV gives the case studies, and Section V draws the conclusions.

II. MATHEMATICAL FORMULATION OF THE PROPOSED MODEL

A. Terms and Assumptions

Urban areas are always sliced into small blocks by streets, alongside which the power feeders can only be laid. For clarity, similar to those mentioned in [9], some terms are explained as below. A *junction* is the crossroad for two streets. A *segment* is the road section between two neighboring junctions. A *block* is an area surrounded by several segments and the customers in it are supplied by one or more distribution transformers, as shown in Fig. 1. Based on on-site investigation, some assumptions are made and described below. 1) Loads are distributed in blocks, and they are supplied by one or more distribution transformers located in the segment around the block. 2) A segment cannot be sliced anymore. If a substation is located in a segment, it will be equivalent to the nearest junction connected to this segment. 3)

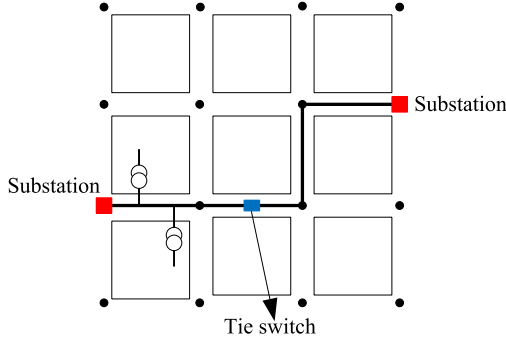


Fig. 2. An illustrative example of single-connection mode.

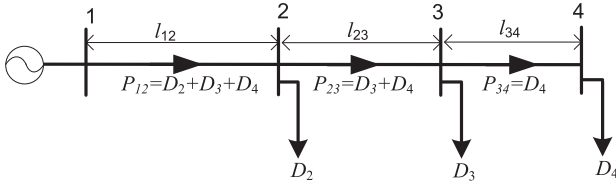


Fig. 3. An illustrative example of the four-node distribution system.

For unification, all feeders share the same type and size. Therefore, only the routing and the number of feeders in the segment need to be optimized. However, the model can be easily extended to consider different types of feeders. 4) For connection mode of feeders, a fixed connection mode, called single-connection mode [22], is considered. Under this condition, the two ends of a feeder are connected to two different substations, and a feeder does not have bifurcation. One tie switch in this feeder will be opened during normal operation condition, and the feeder will be radially operated. Then the “closed loop design, open loop operation” principle is guaranteed by single-connection mode. An illustrative example of single-connection mode is shown in Fig. 2. 5) The planning of switches and breakers are not considered.

B. The Proposed Planning Model

In the proposed model, the load moment is selected as the objective function. Traditionally, the load moment is computed load by load, and the load moment of a load is equal to the value of this load multiplied by the distance between the electricity source and this load point. Takes a four-node distribution system as an example, and it is shown in Fig. 3.

In Fig. 3, l_{12} represents the length of the segment between node 1 and node 2. D_1 represents the value of load connected on node 1. The total load moment is equal to

$$\begin{aligned} & l_{12} \cdot D_2 + (l_{12} + l_{23}) \cdot D_3 + (l_{12} + l_{23} + l_{34}) \cdot D_4 \\ &= l_{12} \cdot (D_2 + D_3 + D_4) + l_{23}(D_3 + D_4) + l_{34}D_4 \\ &= l_{12} \cdot P_{12} + l_{23} \cdot P_{23} + l_{34} \cdot P_{34} \end{aligned} \quad (1)$$

Therefore, it can be appreciated that the total load moment which is originally computed load by load, can also be computed segment by segment, and it is equal to the summation of the

products of the length of a segment and the associated power flow in this segment. In the paper, the power flows in and flows out of a segment could be different, therefore the average power flow in a segment is applied, and it is computed by the summation of power flows in and power flows out over two. Furthermore, considering the possible bi-direction of flow in a segment, the absolute value of power flow is applied. Finally, the objective function of the proposed model is as follows:

$$\min \sum_{s=1}^{NS} \sum_{l=1}^{NL_s} L_s \cdot \left[\sum_{i=start(s)} \sum_{j=end(s)} \left(\frac{|P_{i,j,l}| + |P_{j,i,l}|}{2} \right) \right] \quad (2)$$

In (2) L_s represents the length of segment s . $P_{i,j,l}$ represents the power flows from junction i to junction j in line l . $(|P_{i,j,l}| + |P_{j,i,l}|)/2$ represents the average power flow in line l in segment s which is located between junction i and junction j . Here $i = start(s)$ represents that the start node of segment s is junction i , and $j = end(s)$ represents that the end node of segment s is junction j .

The objective function is subject to the following constraints.

1) *Load Constraints:* Loads in a block are aggregated as an equivalent load. The equivalent load is located in an area and it cannot be directly used in nodal power balance constraint. The equivalent load will be transferred from area to segment, and the corresponding constraints are as follows.

$$\sum_{s=1}^{NS} D_{b,s}^{segment} = D_b^{block}, \exists H_{b,s} = 1, \forall b \quad (3)$$

$$D_{b,s}^{segment} = 0, \exists H_{b,s} = 0, \forall b, s \quad (4)$$

$$\sum_{l=1}^{NL_s} D_{b,s,l}^{segment} = D_{b,s}^{segment}, \forall b, s \quad (5)$$

$$\begin{aligned} P_{i,j,l} + P_{j,i,l} &= \sum_{b=1}^{NB} D_{b,s,l}^{segment}, \forall l, s, \\ i &= start(s), j = end(s) \end{aligned} \quad (6)$$

$$D_{b,s,l}^{segment} \geq 0, \forall b, s, l \quad (7)$$

Expression (3) represents that the load in a block is transferred to those segments which just around this block. Expression (4) represents that if a segment does not around a block, the load in this block will not be transferred to this segment. Expression (5) represents that the load transferred to a segment is supplied by all feeders in this segment. Expression (6) represents the power balance constraints in a segment. The total power flows in a segment from both sides is used to supply all the equivalent loads transferred to this segment. Expression (7) represents the equivalent demand transferred from block b to feeder l of segment s should be non-negative. From expressions (3)-(7), the relationship between block load D_b^{block} and feeder power flow $P_{i,j,l}$ is established. To facilitate the understanding of this relationship, an indicative figure is introduced and it is shown in Fig. 4.

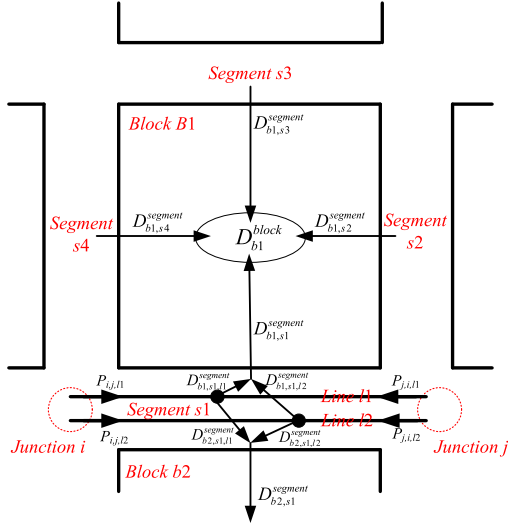


Fig. 4. An indicative figure to show the relationship between block load and feeder power flow.

2) *Substation Output Power Limit Constraints:* In the model, substations are regarded as generation resources. The total power flowed in feeders which are connected to a substation is limited by the maximum allowable output power of this substation multiplied by a utilization factor.

$$\sum_{s=1, \text{start}(s)=ior, \text{end}(s)=i}^{NS} \sum_{l=1}^{NL_s} P_{i,l}^G \leq \eta_i^G \cdot P_i^{Gmax}, \forall i \quad (8)$$

The maximum allowable output power of a substation P_i^{Gmax} is determined by the capacity of the substation. If there is no substation connects to junction j , P_j^{Gmax} takes 0.

3) *Maximum Number of Outgoing Feeder Constraints:* Since the typical number of feeders leaving a substation is limited [1], these constraints can be shown as follows.

$$N_i^{minGL} \leq \sum_{s=1, \text{start}(s)=ior, \text{end}(s)=i}^{NS} \sum_{l=1}^{NL_s} b_{s,l} \leq N_i^{maxGL}, \forall i \quad (9)$$

Constraint (9) represents that the number of all feeders connected to a substation must be smaller than the maximum number of outgoing feeders and larger than the minimum number. If there is no substation located at junction i , this constraint will be relaxed by assigning a large enough value for N_i^{maxGL} and zero for N_i^{minGL} .

4) *Power Balance Constraint for Each Junction:*

$$P_{i,l}^G = \sum_{s=1, \text{and}[\text{start}(s)=i, \text{end}(s)=j]}^{NS} P_{i,j,l} + \sum_{s=1, \text{and}[\text{end}(s)=i, \text{start}(s)=k]}^{NS} P_{i,k,l}, \forall i, l \quad (10)$$

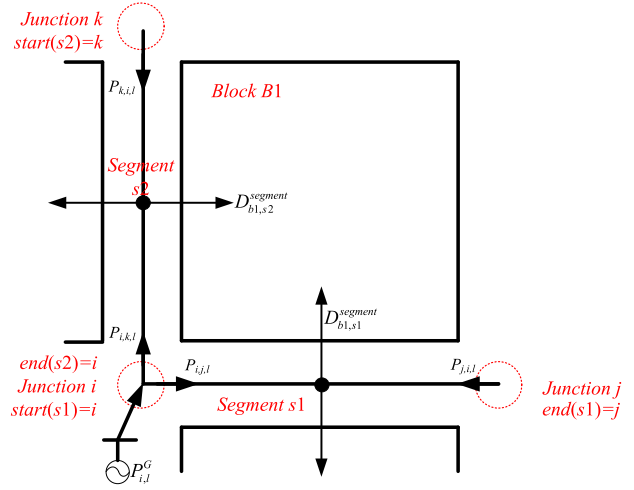


Fig. 5. An indicative figure to facilitate the understanding of power balance constraint at junction i .

For each junction, the power flows in should be equal to the power flows out for each feeder. Here the load is not directly involved since it has been involved in the feeder power flow. To facilitate the understanding of the power balance constraint, an indicative figure is shown in Fig. 5.

5) *Power Flow Limits in Each Segment for Each Feeder:*

$$-P_{s,l}^{max} \cdot \eta_l^L \cdot b_{s,l} \leq P_{i,j,l} \leq P_{s,l}^{max} \cdot \eta_l^L \cdot b_{s,l}, \forall s, l, \quad i = \text{start}(s) \text{ and } j = \text{end}(s) \quad (11)$$

Constraints (11) represent that if a feeder is constructed, the power flow in this feeder is constrained by the capacity of this feeder.

6) *Maximum Number of Feeders Allowed to Be Built in a Segment:*

$$\sum_{l=1}^{NL_s} b_{s,l} \leq N_s^{maxL}, \forall s \quad (12)$$

Constraint (12) represents that the number of feeders allowed to be built in a segment must be less than the given value.

7) *Power Supply Radius Constraint:*

$$\left(\sum_{s=1}^{NS} L_s \cdot b_{s,l} \right) / 2 \leq L^{max_radius}, \forall l \quad (13)$$

The power supply radius of a feeder represents the distance between the substation and the farthest load away from the substation. The power supply radius of a feeder is an indirect indication for feeder voltage drop. This constraint belongs to a relatively soft constraint, and it can be approximated expressed by that the half length of a feeder is smaller than the given maximum power supply radius. Besides, power supply radius constraint coincides with the objective function of minimizing the load moment, and it can be relaxed during optimization.

8) *Feeder Connection Mode Constraints*: In this paper, only the single-connection mode is considered. The single-connection mode can be satisfied if the following two requirements are satisfied. a) There is no bifurcation for each feeder on each junction. b) The two ends of a feeder must be connected to two different substations. Requirement a) can be analytically expressed in (14). Expression (14) means that the degree of each junction node is smaller than two for each feeder.

$$\sum_{s=1, \text{start}(s)=i}^{NS} b_{s,l} \leq 2, \forall l, i \quad (14)$$

When requirement a) is satisfied, requirement b) can be satisfied if the N-1 security criterion of substation is satisfied, which means that when a substation occurs outage, the power supply for each consumer should be guaranteed. The N-1 security criterion of substation can be expressed as follows.

$$\begin{cases} \sum_{s=1, \text{start}(s)=i, \text{end}(s)=i}^{NS} \sum_{l=1}^{NL_s} P_{g,i,l}^G \leq P_i^{G \max}, \forall g, i, L_g^{\text{location}} \neq i \\ P_{g,i,l}^G = 0, \forall g, i, l, L_g^{\text{location}} = i \\ P_{g,i,j,l} + P_{g,j,i,l} = \sum_{b=1}^{NB} D_{b,s,l}^{\text{segment}}, \forall g, l, s, \\ i = \text{start}(s), j = \text{end}(s) \\ P_{g,i,l}^G = \sum_{s=1, \text{and}[\text{start}(s)=i, \text{end}(s)=j]}^{NS} P_{g,i,j,l} \\ + \sum_{s=1, \text{and}[\text{end}(s)=i, \text{start}(s)=j]}^{NS} P_{g,i,k,l}, \forall g, i, l \\ -P_{s,l}^{\max} \cdot b_{s,l} \leq P_{g,i,j,l} \leq P_{s,l}^{\max} \cdot b_{s,l}, \forall g, l, s, \\ i = \text{start}(s) \text{ and } j = \text{end}(s) \\ P_{g,i,j,l} = 0, \forall g, i, j, l, L_g^{\text{location}} = i \end{cases} \quad (15)$$

The outage of a substation is seen as a scenario and expression (15) gives the corresponding constraints under each scenario. Here the variables with subscript g in (15) represents that these variables are correspond to scenario g . The first constraint in (15) represents that if a substation does not occur outage, the nodal power injection caused by this substation is limited by its maximum allowable output power. The second constraint represents that if a substation occurs outage, the nodal power injection caused by this substation is zero. The third constraint represents that the load is supplied by the power flowed in feeders under each scenario. The fourth constraint in (15) represents the nodal power balance under each scenario. The fifth constraint represents the power flow limits under each scenario. The final constraint represents that if a substation occurs outage, for each feeder which connects to this substation, the power flow into the substation or flow out from this substation should be zero.

Due to the characteristics of the single-connection mode of feeders, when the N-1 criterion of substation is considered, the N-1 criterion of feeder will be automatically satisfied. Therefore, the constraints related to N-1 criterion of feeder do not need to be involved in the model.

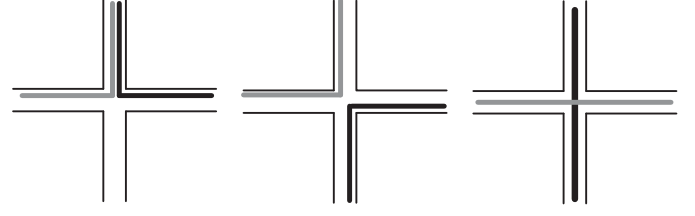


Fig. 6. Possible routing modes of two feeders when they are not in parallel.

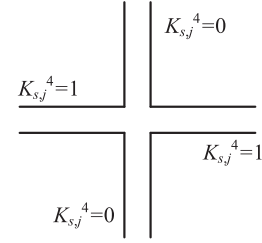


Fig. 7. The value of $K_{s,j}^4$ for the four segments around a junction.

9) *Non-Crossing Constraint of Feeders*: During investigation, the electric power company strongly claims on the non-crossing requirement of feeders. For a junction which connects with three segments, the non-crossing requirement is automatically satisfied. For a junction which connects with four segments, for any two feeders which pass through junction j , if they are in parallel, the non-crossing requirement is automatically satisfied. If they are not in parallel, all the possible routing modes are shown in Fig. 6 based on exhaustive method. It can be found that only the third routing mode will cause feeder crossing.

In order to analytically express the non-crossing requirement, a parameter $K_{s,j}^4$ is introduced and it represents the connection relationship between segment s and junction j which connects with four segments. If segment s is not connected to junction j or junction j is not connected with four segments, $K_{s,j}^4$ will take 0. If segment s is connected to junction j and junction j is connected with four segments, the four segments which around the junction will be assigned the value of 0, 1, 0, 1, respectively. An illustrative figure is shown in Fig. 7.

For any two feeders $l1$ and $l2$ which pass through junction j , they will satisfy the non-crossing requirement if they satisfy the following constraint.

$$\begin{aligned} 1 &\leq \sum_{s=1, j=\text{start}(s) \text{ or } j=\text{end}(s)}^{NS} K_{s,j}^4 \cdot b_{s,l1} + \sum_{s=1, j=\text{start}(s) \text{ or } j=\text{end}(s)}^{NS} (1 - K_{s,j}^4) \\ &\quad \cdot b_{s,l2} \leq 3, \\ &\forall j \in \Omega_4, l1, l2, l1 \neq l2, \\ &\exists \left(\sum_{s=1, j=\text{start}(s) \text{ or } j=\text{end}(s)}^{NS} b_{s,l1} = 2 \text{ and } \sum_{s=1, j=\text{start}(s) \text{ or } j=\text{end}(s)}^{NS} b_{s,l2} = 2 \right) \end{aligned} \quad (16)$$

where Ω_4 represents the set of junctions which connect with four segments. Expression (16) can be explained as follows. Since $K_{s,j}^4 \cdot b_{s,l1}$ and $(1 - K_{s,j}^4) \cdot b_{s,l2}$ are all binary variables, and if a feeder passes through a junction, it will pass through two segments around this junction, then the value of the middle term of (16) would be 0, 1, 2, 3, 4. If both $l1$ and $l2$ are planned to be constructed and they pass through two segments around junction j , based on exhaustive method, it can be appreciated that only when $l1$ and $l2$ are crossed, the value of the middle term in (16) will take the value of 0 or 4. Therefore, expression (16) can guarantee the non-crossing requirement of feeders. For a junction with more than four segments, the corresponding non-crossing constraints are introduced in the Appendix.

C. Improvement of the Model

1) *Linearization of the Objective Function*: Since the absolute sign exists in the objective function and the objective function is minimized, the objective function of (1) can be equally linearized as follows.

$$\begin{cases} \min \sum_{s=1}^{NS} \sum_{l=1}^{NL_s} L_s \cdot \left[\sum_{i=start(s)} \sum_{j=end(s)} \left(\frac{P_{i,j,l}^{abs} + P_{j,i,l}^{abs}}{2} \right) \right] \\ P_{i,j,l}^{abs} \geq P_{i,j,l} \\ P_{i,j,l}^{abs} \geq -P_{i,j,l} \\ P_{j,i,l}^{abs} \geq P_{j,i,l} \\ P_{j,i,l}^{abs} \geq -P_{j,i,l} \end{cases} \quad (17)$$

where $P_{i,j,l}^{abs}$ and $P_{j,i,l}^{abs}$ are auxiliary variables.

After the objective function is linearized, the entire optimization model belongs to a MILP problem and it can be solved by state-of-art commercial solver.

2) *Elimination of Redundant Feeders*: Furthermore, from expression (2) or (17), it can be appreciated that zero power flow feeders will not affect the value of the objective function. During optimization it is found that redundant feeders, in which the feeder flow is zero, would be planned, and this phenomenon makes the planning results deviate from the desired one. In order to avoid the construction of redundant feeders, an additional constraint is added and this constraint is shown as follows.

$$b_{s,l} \leq \frac{P_{i,j,l}^{abs} + P_{j,i,l}^{abs}}{\min_D_b^{block}}, \forall l, s, \exists [start(s) = i \text{ and } end(s) = j] \quad (18)$$

Here $\min_D_b^{block}$ is the minimum block load among all block loads. It can be appreciated that for each feeder, the absolute values of the power which flow in a segment from both sides should be larger than any block loads, except for redundant feeders. If the right-hand side of (18) is small than one, the feeder belongs to redundant feeder and it should not be planned. Therefore expression (18) can help to screen the redundant feeders out of the total feeders.

3) *Simplification of the Non-Crossing Constraints*: From expression (16), it can be found that if no feeder is constructed in the segment which connects to junction j , the value of the middle term of (16) will be 0. If only one feeder is constructed, the value of the middle term of (16) will take 0 or 1 or 2. Therefore, the

left inequality constraint can be reformulated as below.

$$\begin{aligned} & \sum_{s=1, start(s)=j}^{NS} K_{s,j}^4 \cdot b_{s,l1} + \sum_{s=1, start(s)=j}^{NS} (1 - K_{s,j}^4) \cdot b_{s,l2} \\ & \geq \sum_{s=1, start(s)=j}^{NS} \frac{b_{s,l1}}{2} + \sum_{s=1, start(s)=j}^{NS} \frac{b_{s,l2}}{2} - 1, \\ & \forall j \in \Omega_4, l_1, l_2, l_1 \neq l_2; \end{aligned} \quad (19)$$

It can be appreciated that if there is less than one feeder is constructed in the segment which connects to junction j , the right-hand side of (19) will be equal to or smaller than 0, and this constraint will be automatically satisfied. If two feeders are constructed, the right-hand side of (19) will be 1, and (19) will be equivalent to the left inequality constraint of (16).

If the number of constructed feeders which pass through junction j is smaller than or equal to 1, the value of the middle term of (16) will take 0 or 1 or 2. Under this condition, the right inequality constraint of (16) is automatically be satisfied. The right inequality constraint of (16) can be equivalently expressed as below.

$$\begin{aligned} & \sum_{s=1, start(s)=j}^{NS} K_{s,j} \cdot b_{s,l1} - \sum_{s=1, start(s)=j}^{NS} (1 - K_{s,j}) \cdot b_{s,l2} \\ & \leq 3, \forall j \in \Omega_4, l_1, l_2, l_1 \neq l_2; \end{aligned} \quad (20)$$

III. CASE STUDY

The proposed model is tested on a 9-block test system and a real 22-block district distribution system in Shandong province of China. The optimization model is solved by a MILP solver of CPLEX [25] on GAMS platform [26]. All computations are performed on a computer with a 3.7G Hz CPU and 64G memory.

A. Case Studies of the 9-Block Test System

The 9-block system contains 9 blocks, 16 junctions, 24 segments and 2 substations. The relationship between blocks, segments, and junctions are shown in Fig. 8. The two substations are located on junction 2 and junction 15, respectively, and they are marked by red circles. The forecast load of each block is 4 MW. The length of each segment is 0.5 km. The maximum power supply radius is 3 km. The maximum number of feeders allowed to be built in a segment is 4. The maximum and minimum numbers of outgoing feeders of substations are 15, and 0, respectively. The maximum allowable power flows in feeders is 10 MW. The maximum allowable output power of each substation is 50 MW. The utilization factor of substation is 80%.

After optimization, the planned feeders are described by different colors and they are shown in Fig. 9. From Fig. 9 it can be found that four feeders are built. The feeder power flow in each segment is also shown, and how the block loads are supplied can also be observed. The computation time is about 3 seconds. In

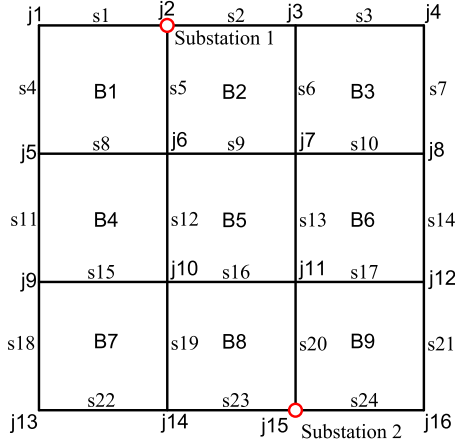


Fig. 8. The hypothetical 9-block system.

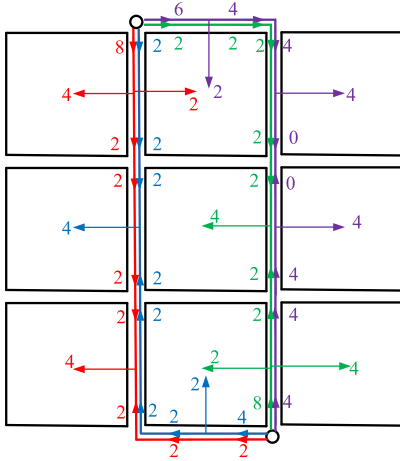


Fig. 9. Planning result of the 9-block system.

terms of feeder routing, it can be found that the feeders satisfy the single-connection mode, and they are not crossed. In terms of given power flow, it can be found that the block loads tend to be equally supplied by four feeders. It can also be found that the planned feeders can withstand the single outage of a substation or a feeder in any segment.

B. Case Studies of the 22-Block Test System

The 22-block district distribution system is shown in Fig. 10. This system contains 22 blocks, 36 junctions, 57 segments and 4 substations. The four substations are located on junction 1, 5, 9 and 34, respectively, and they are marked by red circles. The forecast block load of each is shown in Table I. The length of each segment is shown in Table II. The maximum power supply radius is 4 km. The maximum number of feeders allowed to be built in a segment is 3. The maximum and minimum numbers of outgoing feeders of the substations are 15, and 0, respectively. The maximum allowable power flows in feeder is 10 MW. The maximum allowable output power of the substation is 50 MW. The utilization factor of substation is 80%. The utilization factor of feeder is 100%.

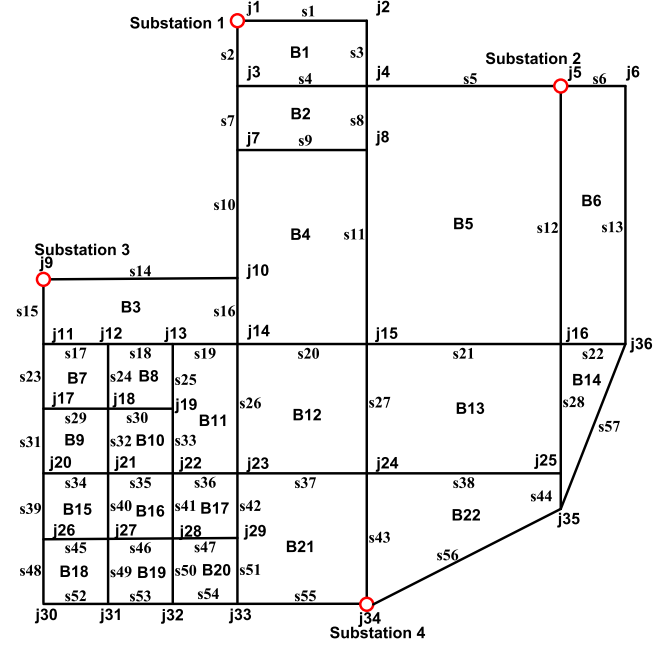


Fig. 10. Topology of the 22-block system.

TABLE I
LOADS IN EACH BLOCK

Index of Block	Load (MW)	Index of Block	Load (MW)	Index of Block	Load (MW)
B1	2.693	B9	0.724	B17	1.007
B2	0.918	B10	0.659	B18	0.801
B3	3.858	B11	2.215	B19	1.404
B4	3.499	B12	2.754	B20	1.705
B5	5.075	B13	3.725	B21	3.469
B6	3.629	B14	3.091	B22	2.929
B7	0.712	B15	0.738		
B8	0.641	B16	1.896		

After optimization, the optimal feeder routing and number of feeders are shown in Fig. 11. A post-calculation is applied to compute the length of each feeder, and the length of each feeder is shown in Table III. It is found that the power supply radius constraint is satisfied.

From Fig. 11 it can be found that seven feeders should be built. The directions of the feeder power flow in each segment are described by arrows. For each line, the power flows from the substation to the terminal node, and a zero power flow in the figure implies the location of a terminal node. And how the block loads are supplied by different feeders is also shown in the figure. It can be found that block 4, 5, 6, 14, 17, 18, and 22 are supplied by more than one transformer.

The computation time is about 83 minutes using a pre-specified duality gap of 0.1%. In the model, the most important variable is $b_{s,l}$, which represents whether feeder is constructed or not. The binary variable $b_{s,l}$ is relatively robust with respect to the selected duality gap. For example, when the duality gap takes 0.5%, the obtained binary variable $b_{s,l}$ is the same as that obtained by the model with duality gap of 0.1%. Therefore, a relatively larger duality gap, i.e., 0.5%, can be applied in the

TABLE II
LENGTH OF EACH SEGMENT

Index	Length (km)	Index	Length (km)	Index	Length (km)	Index	Length (km)
s1	0.766	s16	0.751	s31	0.406	s46	0.299
s2	1.249	s17	0.328	s32	0.403	s47	0.557
s3	1.192	s18	0.303	s33	0.406	s48	0.518
s4	0.694	s19	0.579	s34	0.324	s49	0.498
s5	1.631	s20	0.714	s35	0.289	s50	0.521
s6	1.243	s21	1.528	s36	0.552	s51	0.557
s7	0.299	s22	1.261	s37	0.701	s52	0.337
s8	0.157	s23	0.337	s38	1.513	s53	0.306
s9	0.743	s24	0.329	s39	0.341	s54	0.577
s10	0.293	s25	0.316	s40	0.355	s55	0.656
s11	1.221	s26	0.736	s41	0.348	s56	1.773
s12	1.313	s27	0.718	s42	0.357	s57	1.673
s13	1.298	s28	0.685	s43	0.959		
s14	1.316	s29	0.326	s44	0.237		
s15	0.363	s30	0.279	s45	0.322		

TABLE III
LENGTH OF EACH FEEDER

Index of feeder	Route of the feeder	Length of the feeder (km)
1	s2-s7-s10-s14	3.157
2	s1-s3-s8-s11-s27-s43	5.013
3	s15-s23-s31-s39-s45-s49-s53-s54-s55	3.806
4	s15-s23-s29-s32-s35-s36-s42-s51-s55	3.840
5	s15-s17-s18-s25-s33-s36-s37-s43	3.928
6	s12-s28-s44-s56	4.008
7	s12-s28-s44-s56	4.008

For example, the experts may give priority to establish feeders along main streets, or give priority to use the assigned tunnels. The restricted access of some segments, and the isolation caused by railways, canals and rivers could also be respected. Considering the prior knowledge can pre-fix some variables, especially binary variables, which can significantly improve the computation efficiency.

IV. CONCLUSION

In this paper a feeder planning model for the urban distribution system is proposed. "Supply electricity to loads using nearby power sources" is selected as the objective, and it is analytically represented by load moment. Various practical requirements, such as block loads, practical street layout, meshed network configuration, connection mode of feeders, power supply radius, constraint, non-crossing constraint of feeders, maximum number of outgoing feeder constraints, and maximum number of feeders allowed to be built in a segment, etc., are respected. The model is recast as a MILP problem. The proposed model is tested on a 9-block test system and a real 22-block district distribution system. In the future works, more practical requirements will be collected and analytically expressed in the model, and the dilemma between rigid optimization and flexible decision making will be further analyzed. Heuristic methods and approximation methods are required to improve the computation efficiency. The non-crossing requirement will be further analyzed.

APPENDIX

For a junction which connects with m ($m > 4$) segments, in order to analytically express the non-crossing requirement, a parameter $K_{s,j}^m$ is introduced and it reflects the connection relationship between segment s and junction j which connects with m segments. If segment s is not connected to junction j or junction j is not connected with m segments, $K_{s,j}^m$ will take 0. If segment s is connected to junction j and j is connected with m segments, the segments which around junction j in clockwise order will be assigned the value of 1, 2, 3, ..., m , respectively. An illustrative figure to show the value of $K_{s,j}^m$ is shown in Fig. 13.

After analysis it is observed that two feeders will satisfy the non-crossing requirement if both the two segments related to a feeder are just laid within the same space which is surrounded by the two segments related to another feeder. This description includes two conditions and they are illustrated in Fig. 13. In

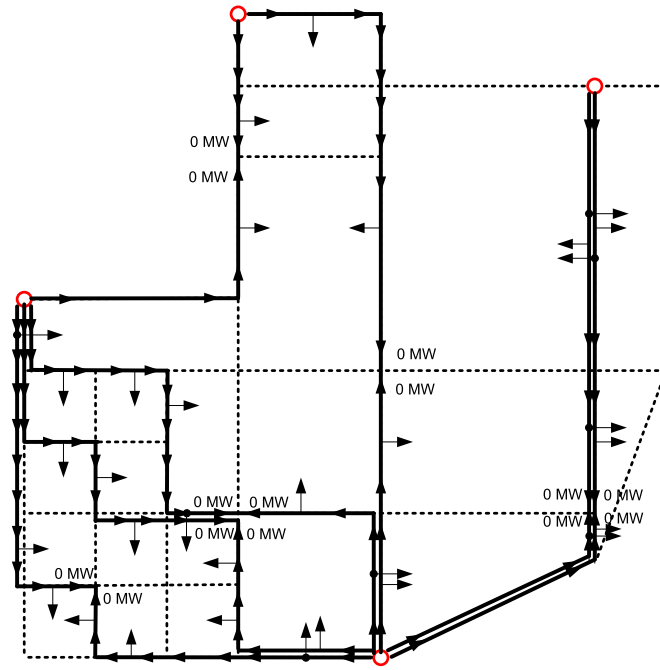


Fig. 11. Planning results of the 22-block system.

model. When the duality gap takes 0.5%, the computation time is about 23 minutes.

For the proposed model, the number of binary variables is equal to $NS \cdot NL_s$. For the 22-block system is concerned, this number is 171. Although the number of binary variables is relatively small, a candidate feeder routing corresponds to a combination of many binary variables, and the number of possible combination is tremendously large, which incurs a heavy computation burden for the proposed model.

The computation efficiency of the proposed model would be improved by many measures. Pre-processing of the planning area can improve the computation efficiency. For example, the adjacent small areas can be integrated and seen as one block, and they can be supplied through auxiliary lines. Besides, prior knowledge can be applied to improve the computation efficiency.

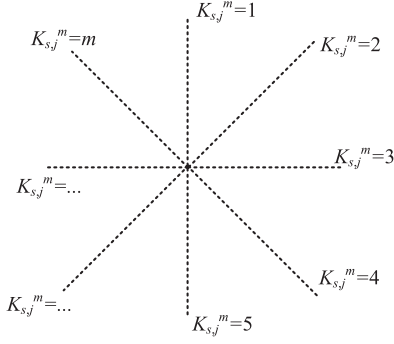


Fig. 12. The value of $K_{s,j}^m$ for the m segments around a junction.

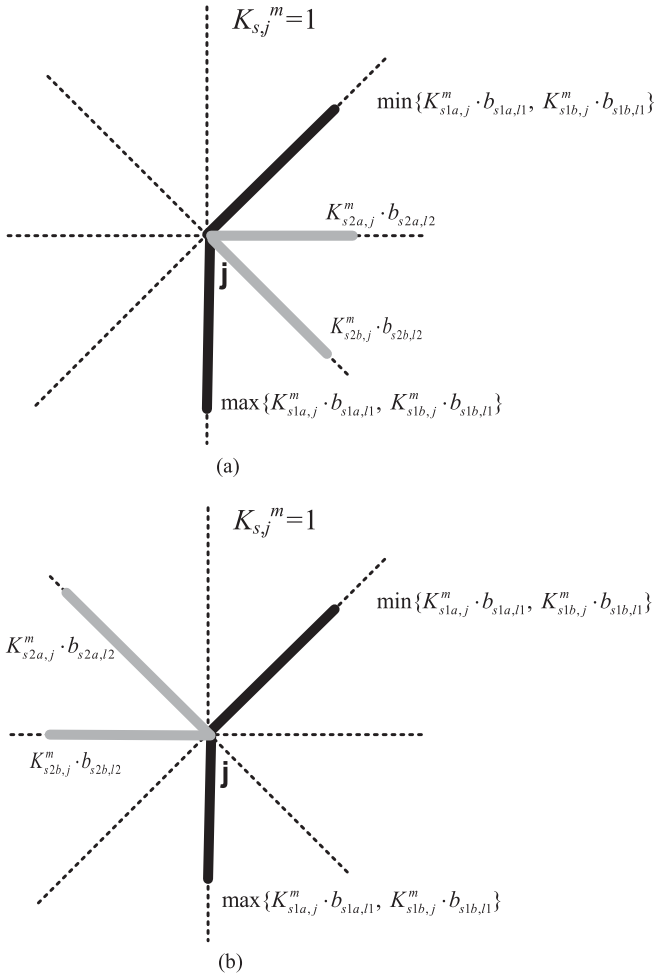


Fig. 13. (a) Two situations correspond to the non-crossing constraint. (b) Two situations correspond to the non-crossing constraint.

Fig. 13, the black and grey lines represent the two feeders which pass through junction j .

For any two feeders $l1$ and $l2$ which pass through junction j , the indices of the two segments related to feeder $l1$ are marked by $s1a, s1b$. The indices of the two segments related to feeder $l2$ are marked by $s2a$, and $s2b$. The non-crossing requirement can

be analytically expressed as below.

$$\begin{cases} \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \leq K_{s2a,j}^m \cdot b_{s2a,l2} \\ \leq \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \\ \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \leq K_{s2b,j}^m \cdot b_{s2b,l2} \\ \leq \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \end{cases}, \quad \exists b_{s1a,l1}, b_{s1b,l1}, b_{s2a,l2}, b_{s2b,l2} = 1 \quad (A1)$$

or

$$\begin{cases} \left\{ \begin{aligned} &K_{s2a,j}^m \cdot b_{s2a,l2} \leq \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \\ &\text{or} \\ &K_{s2a,j}^m \cdot b_{s2a,l2} \geq \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \end{aligned} \right\} \\ \text{and} \\ \left\{ \begin{aligned} &K_{s2b,j}^m \cdot b_{s2b,l2} \leq \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \\ &\text{or} \\ &K_{s2b,j}^m \cdot b_{s2b,l2} \geq \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \end{aligned} \right\} \end{cases}, \quad \exists b_{s1a,l1}, b_{s1b,l1}, b_{s2a,l2}, b_{s2b,l2} = 1 \quad (A2)$$

Here constraint (A1) corresponds to the condition which is shown in Fig. 13(a), while constraint (A2) corresponds to the condition which is shown in Fig. 13(b). The term of $(b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2})$ in constraints (A1)-(A2) can guarantee that only when feeders $l1$ and $l2$ are both planned, constraints (A1)-(A2) will be activated. Otherwise constraints (A1)-(A2) will be relaxed.

Constraints (A1)-(A2) can be further expressed as below.

$$\begin{aligned} &b_{s,l1,l2,j}^{aux1} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\ &\quad \cdot \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \\ &\leq b_{s,l1,l2,j}^{aux1} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\ &\quad \cdot K_{s2a,j}^m \cdot b_{s2a,l2} \\ &\leq b_{s,l1,l2,j}^{aux1} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\ &\quad \cdot \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \end{aligned} \quad (A3)$$

$$\begin{aligned} &b_{s,l1,l2,j}^{aux1} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\ &\quad \cdot \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \\ &\leq b_{s,l1,l2,j}^{aux1} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\ &\quad \cdot K_{s2b,j}^m \cdot b_{s2b,l2} \\ &\leq b_{s,l1,l2,j}^{aux1} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\ &\quad \cdot \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \end{aligned} \quad (A4)$$

$$\begin{aligned} &(1 - b_{s,l1,l2,j}^{aux1}) \cdot b_{s,l1,l2,j}^{aux2} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\ &\quad \cdot K_{s2a,j}^m \cdot b_{s2a,l2} \\ &\leq (1 - b_{s,l1,l2,j}^{aux1}) \cdot b_{s,l1,l2,j}^{aux2} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\ &\quad \cdot \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \end{aligned} \quad (A5)$$

$$\begin{aligned}
& (1 - b_{s,l1,l2,j}^{aux1}) \cdot (1 - b_{s,l1,l2,j}^{aux2}) \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \\
& \quad \cdot b_{s2b,l2}) \cdot K_{s2a,j}^m \cdot b_{s2a,l2} \\
\geq & (1 - b_{s,l1,l2,j}^{aux1}) \cdot (1 - b_{s,l1,l2,j}^{aux2}) \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \\
& \quad \cdot b_{s2b,l2}) \cdot \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \quad (A6)
\end{aligned}$$

$$\begin{aligned}
& (1 - b_{s,l1,l2,j}^{aux1}) \cdot b_{s,l1,l2,j}^{aux3} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\
& \quad \cdot K_{s2b,j}^m \cdot b_{s2b,l2} \\
\leq & (1 - b_{s,l1,l2,j}^{aux1}) \cdot b_{s,l1,l2,j}^{aux3} \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \cdot b_{s2b,l2}) \\
& \quad \cdot \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \quad (A7)
\end{aligned}$$

$$\begin{aligned}
& (1 - b_{s,l1,l2,j}^{aux1}) \cdot (1 - b_{s,l1,l2,j}^{aux3}) \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \\
& \quad \cdot b_{s2b,l2}) \cdot K_{s2b,j}^m \cdot b_{s2b,l2} \\
\geq & (1 - b_{s,l1,l2,j}^{aux1}) \cdot (1 - b_{s,l1,l2,j}^{aux3}) \cdot (b_{s1a,l1} \cdot b_{s1b,l1} \cdot b_{s2a,l2} \\
& \quad \cdot b_{s2b,l2}) \cdot \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \quad (A8)
\end{aligned}$$

Constraint (A1) is reformulated as (A3)-(A4), and constraint (A2) is reformulated as (A5)-(A8). Here $b_{s,l1,l2,j}^{aux1}$ is a free auxiliary binary variable. It can guarantee that either constraints (A3)-(A4) or constraints (A5)-(A8) are activated. $b_{s,l1,l2,j}^{aux2}$ is also a free auxiliary binary variable. It can guarantee that either constraint (A5) or (A6) is activated. Similarly, the auxiliary binary variable $b_{s,l1,l2,j}^{aux3}$ can guarantee that either constraint (A7) or (A8) is activated.

In constraints (A3)-(A8), the expressions of $\min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\}$ and $\max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\}$ can be equivalently expressed as follows.

$$\begin{aligned}
& \min\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \\
& = (K_{s1a,j}^m \cdot b_{s1a,l1}) \cdot (1 - b_{s,l1,j}^{aux4}) + (K_{s1b,j}^m \cdot b_{s1b,l1}) \cdot b_{s,l1,j}^{aux4} \quad (A9)
\end{aligned}$$

$$\begin{aligned}
& \max\{K_{s1a,j}^m \cdot b_{s1a,l1}, K_{s1b,j}^m \cdot b_{s1b,l1}\} \\
& = (K_{s1a,j}^m \cdot b_{s1a,l1}) \cdot b_{s,l1,j}^{aux4} + (K_{s1b,j}^m \cdot b_{s1b,l1}) \cdot (1 - b_{s,l1,j}^{aux4}) \quad (A10)
\end{aligned}$$

where the auxiliary binary variable $b_{s,l1,j}^{aux4}$ satisfies

$$b_{s,l1,j}^{aux4} = \begin{cases} 1, & \exists K_{s1a,j}^m \cdot b_{s1a,l1} \geq K_{s1b,j}^m \cdot b_{s1b,l1} \\ 0, & \exists K_{s1a,j}^m \cdot b_{s1a,l1} < K_{s1b,j}^m \cdot b_{s1b,l1} \end{cases} \quad (A11)$$

And (A11) can be equivalently linearized as below.

$$\begin{aligned}
& \frac{K_{s1a,j}^m \cdot b_{s1a,l1} - K_{s1b,j}^m \cdot b_{s1b,l1}}{M} \leq b_{s,l1,j}^{aux4} \leq 1 \\
& + \frac{K_{s1a,j}^m \cdot b_{s1a,l1} - K_{s1b,j}^m \cdot b_{s1b,l1}}{M} \quad (A12)
\end{aligned}$$

Here M is a large enough positive number.

For each term in (A3)-(A8), it is expressed as the product of some binary variables and a parameter, and the product of some binary variables can be equivalently linearized [27]. Then the entire non-crossing requirement can be linearized.

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Mingqiang Wang (Member, IEEE) received the Ph.D. degree from Nanyang Technological University, Singapore, in 2012. He is currently an Associate Professor with the School of Electrical Engineering, Shandong University, Jinan, China. His research interests include power system economic operation and control, and planning.



Ming Yang (Senior Member, IEEE) received the B.Eng. and Ph.D. degrees in electrical engineering from Shandong University, Jinan, China, in 2003 and 2009, respectively. He is currently a Professor with Shandong University. His research interests include power system optimal operation and control. He is an Associate Editor for IEEE TRANSACTIONS ON POWER SYSTEMS, IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS, and *IET Renewable Power Generation*.



Zhen Fang received the B.S. degree in electrical engineering in 2019 from the School of Electrical Engineering, Shandong University, where he is currently working toward the master's degree with the School of Electrical Engineering. His research interests include renewable energy integration and power system optimal operation.



Mengxia Wang (Member, IEEE) received the Ph.D. degree in electrical engineering from Shandong University, Jinan, China, in 2011. From July 2011 to July 2013, he conducted postdoctoral research with the Department of Electrical Engineering, Tsinghua University, Beijing, China. He is currently an Associate Professor with Shandong University. His research interests include power system security analysis, optimal operation, and control.



Qiuwei Wu (Senior Member, IEEE) received the Ph.D. degree in power system engineering from Nanyang Technological University, Singapore, in 2009. Since 2009, he has been an Assistant Professor with the Department of Electrical Engineering, Technical University of Denmark, Kongens Lyngby, Denmark. His research interests include operation and control of power systems with high penetration of renewables, including wind power modeling and control, active distribution networks, and operation of integrated energy systems. He is the Deputy-Editor-in-Chief of the *International Journal of Electrical Power and Energy Systems*. He is the Editor of IEEE TRANSACTIONS ON SMART GRID and IEEE POWER ENGINEERING LETTERS. He is the Regional Editor for Europe of *IET Renewable Power Generation*, a Subject Editor of *IET Generation, Transmission and Distribution*, and an Associate Editor for the *Journal of Modern Power Systems and Clean Energy*.